

Raphael Karamagi

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EDUCATION

Duke University

Aug. 2025 – May 2029

B.S.E. Electrical & Computer Engineering, B.S. Computer Science, GPA: 3.83/4.00

Durham, NC

Selected Coursework: Computer Architecture, Data Structures & Algorithms, Signals & Systems, Probability, Linear Algebra, Multivariable Calculus

EXPERIENCE

Duke University OIT

Aug 2026 – Present

Student Developer – Innovation Co-Lab

Durham, NC

- Selected to join Duke OIT's developer team for the 2026–27 academic year, supporting internal tool & infrastructure development and technical workshops for the Duke community

Duke University

May 2026 – July 2026

Software Engineer Intern – Code+ Program

Durham, NC

- Built an ML model security & safety scanning pipeline—spanning model weights, dependencies, and inference red-teaming—producing risk-scored reports before 50+ ML models are deployed to 20,000+ Duke faculty and students.
- Deployed as a containerized platform backed by Postgres, GitLab CI pipelines, and OIDC user authentication.

IntelliSOFT Consulting

July 2024 – Aug. 2024

Software Engineer Intern

Nairobi, Kenya

- Implemented REST API endpoints with role-based access in OpenMRS EMR; led deployment to 20+ hospitals across South Sudan and 3 Nairobi pilot sites with 98% uptime.

PROJECTS

Stock Prediction Platform v2 | Python, PyTorch, JavaScript, C++, FinBERT

Jan. 2026 – Present

- Led a 5-person team building a dual-model (LSTM + NLP) stock direction prediction system across 20 tickers; achieved 57% and 63% test accuracy on next day movement & volatility respectively.
- Currently architecting v2 around an event-driven backtester with order book simulation, transaction cost modeling, and walk-forward validation

Compost Now Wash Automation | Arduino, C/C++, MOSFET

Jan. 2026 – Apr. 2026

- Led team designing automated water shutoff for industrial bin washers, reducing 200gal of wasted water for the 1000+ bins washed per day.
- Designed embedded control circuit (Arduino Nano + 12V solenoid via MOSFET switches, LED beacons, user inputs) and wrote non-blocking C++ firmware.

ASL Computer Vision Classifier | PyTorch, OpenCV

Oct. 2025 – Dec. 2025

- Built a dual-model CV system classifying 29 ASL gestures from 87K+ images (71% test accuracy); designed Landmark NN (~10K parameters) for real-time, local CPU inference with 100% hand-detection accuracy.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, R, SQL, JavaScript

ML & Data: PyTorch, TensorFlow, scikit-learn, OpenCV, pandas, NumPy

Frameworks & Tools: Flask, React, Node.js, PostgreSQL, Git, Docker, Linux, OAuth

LEADERSHIP & ACTIVITIES

Duke Applied Machine Learning – Senior Engineer

Sep. 2025 – Present

- Mentor junior members through the AITP program; lead Stock Prediction Platform team.

National Society of Black Engineers (NSBE) – Duke Chapter

Aug. 2025 – Present

- Represented Duke at Fall 2025 Regional Conference and 2026 Annual Convention (Baltimore, MD).

HONORS & AWARDS

5th Globally, Cambridge AS-Level Economics | UKMT Senior Math Challenge Silver (2024) | SAT 1540/1600